Project Report

SETU
Module: SEM2 -
Networking: Wireless
and Routing Concepts -
Y2

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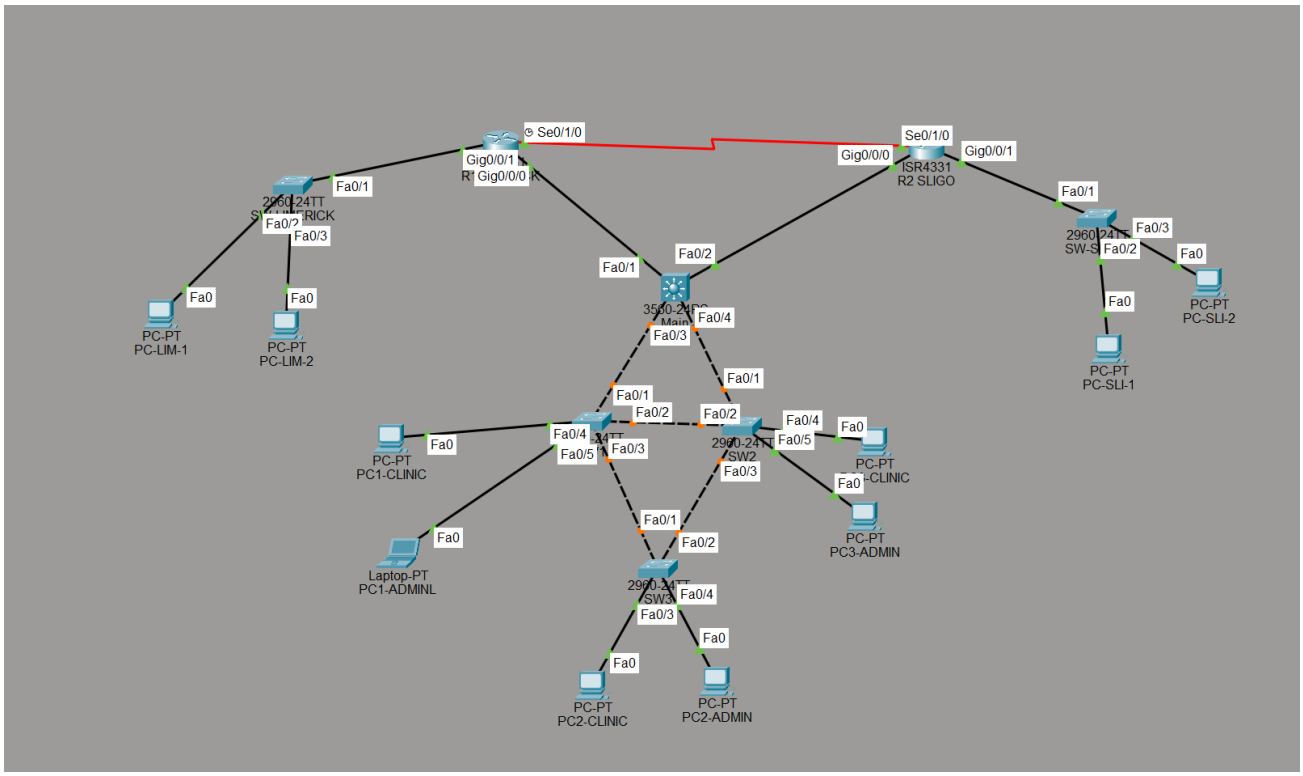
Aurora Healthcare Group: Network Design and Security Implementation Report

1. Introduction

Within this project, a description of a complete and comprehensive design, configuration, and security hardening of a secure and robust LAN-based internetworking solution for a growing Irish-based private healthcare provider, Aurora Healthcare Group, has been provided. The main goal and objective of this project was to engineer a secure and robust internetworking solution, linking a Central Campus location in Dublin to two regional branch clinics located in Limerick and Sligo.

In order to prove and verify the effectiveness and appropriateness of a chosen internetworking solution, **a variety of testing has been carried out**, proving not only that the implemented internetworking solution meets all specified and required criteria and requirements but also exceeds them.

2. Network Topology Overview



- **Central Campus (Dublin):** A Cisco 3560 Multilayer Switch (Main_MLS) serves as the collapsed core/distribution layer device. It handles all inter-VLAN routing, provides centralized DHCP services, and acts as the secure entry point for remote administration. Below the MLS, three Cisco 2960 Layer 2 access switches are deployed in a redundant, looped topology. This loop deliberately invokes the Spanning Tree Protocol (STP) to prevent broadcast storms while providing alternative paths for internal traffic.
- **Regional Clinics (Limerick & Sligo):** Following the requirement for simplicity in remote branches, each clinic operates as a **"flat LAN"** without complex VLAN segmentation. Each site is equipped with a Cisco ISR 4331 Edge Router, which acts as the local DHCP server and gateway, connected to a local Layer 2 switch.

3. Requirements Mapping and Design Choices

Requirement	Design Choice & Configuration
1. VLAN Design	**Central Campus:**Segmented into VLAN 100 (Clinical, 110 users) and VLAN 200 (Admin, 25 users). Inter-VLAN routing was implemented on the Main_MLS using Switched Virtual Interfaces (SVIs) instead of a Router-on-a-Stick, providing wire-speed routing performance which is essential for a large campus. **Clinics:** Left as unsegmented flat LANs to minimize administrative overhead.
2. IP Addressing & DHCP	**Central Campus:** Main_MLS hosts two DHCP pools. VLAN 100 uses 10.10.100.0/25 and VLAN 200 uses 10.10.200.0/27. **Clinics:** Local edge routers handle DHCP to ensure branch autonomy during WAN outages. Limerick uses 172.16.10.0/27, and Sligo uses 172.16.20.0/27.
3. Layer 2 Security	Comprehensive security measures **(Port Security, BPDU Guard, DHCP Snooping, DAI)** were applied to all access switches. Additionally, all unused switch ports were placed in a "Blackhole" VLAN (VLAN 67) and administratively shut down.
4. Secure Remote Admin	SSH version 2 was configured on Main_MLS with a **local domain** (aurora.ie) and **1024-bit RSA keys**. Telnet was explicitly blocked. Two local accounts were created: admin (Privilege Level 15) and labuser (Privilege Level 5).
5. Static & Floating Routing	Full mesh reachability was achieved using static routes. Primary routes point to the Main_MLS (Default Administrative Distance of 1). Floating static routes were configured across the serial link with an Administrative Distance of 50, ensuring they remain inactive until primary link failure.

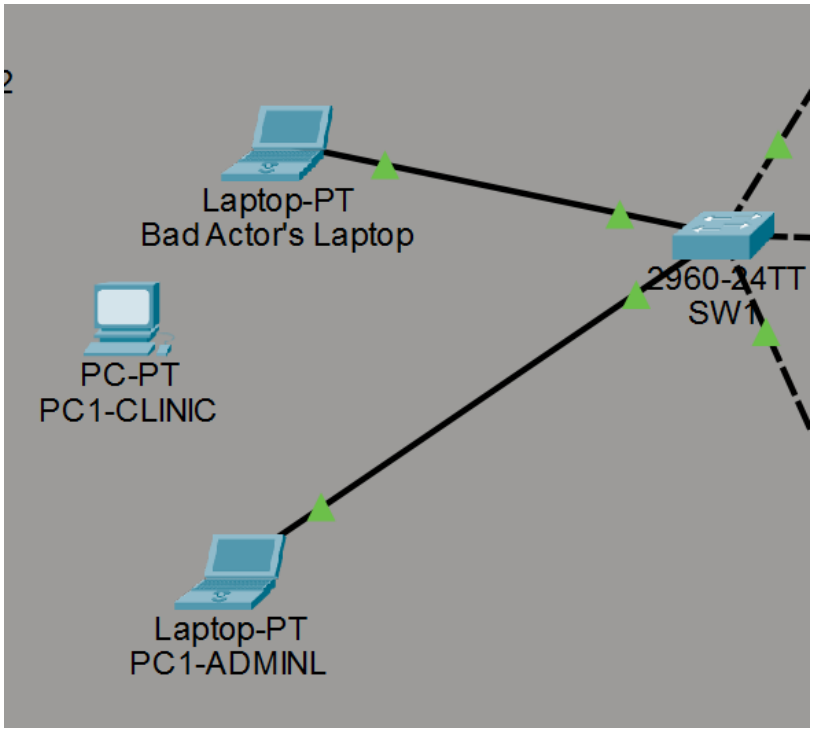
4. LAN and Layer 2 Security Hardening (With Proof of Concept)

Protecting the Central Campus LAN from internal threat vectors was a critical focus of this design. The following mitigation strategies were implemented and tested:

4.1. Mitigating MAC Table and VLAN Hopping Attacks

To prevent an attacker from flooding the switch's MAC address table (CAM table overflow), **Port Security** was enabled on all end-user ports. The ports are restricted to a maximum of two dynamically learned (sticky) MAC addresses. If a third unauthorized device is connected, the port enters restrict mode, dropping unauthorized frames while generating log alerts without completely shutting down the port. Furthermore, Dynamic Trunking Protocol (DTP) was disabled (switchport nonegotiate), and ports were hardcoded to access mode to prevent VLAN hopping attacks.

Proof of Concept: When a rogue laptop was connected to an authorized port on SW1, the switch immediately restricted the traffic.



SW1

Physical Config CLI Attributes

```
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address
0006.2A88.2D3D on port FastEthernet0/4.
%PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address
0006.2A88.2D3D on port FastEthernet0/4.
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%PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address
0006.2A88.2D3D on port FastEthernet0/4.


User Access Verification

Password:

SW1>enable
Password:
SW1#show port-security interface fa0/4
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Restrict
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 2
Total MAC Addresses        : 2
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 2
Last Source Address:Vlan   : 0006.2A88.2D3D:100
Security Violation Count   : 9

SW1#
```

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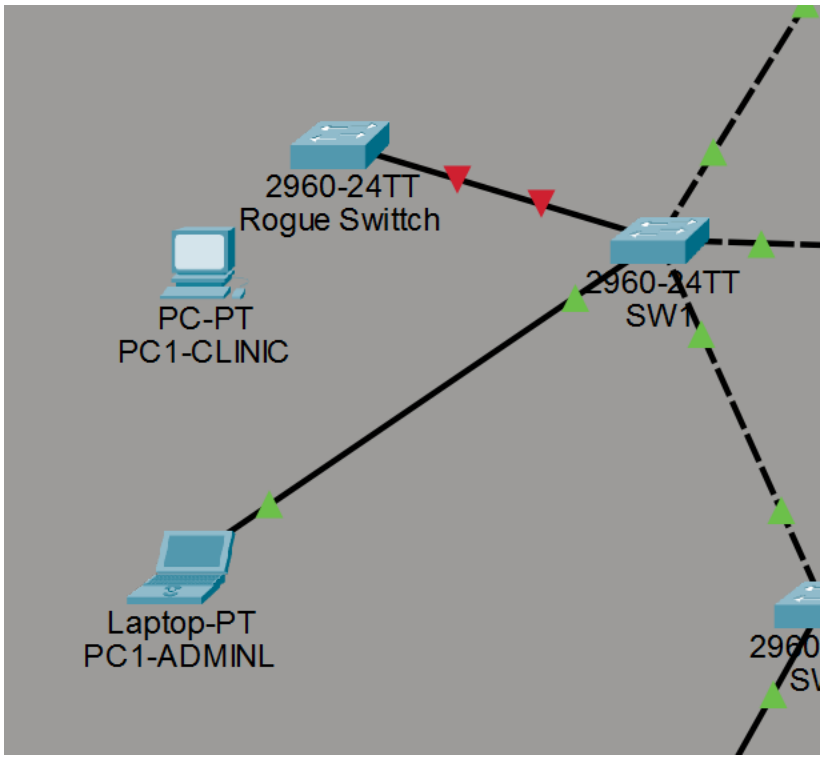
4.2. Safeguarding the Spanning Tree Topology (STP Attacks)

Because the Central Campus switches are wired in a loop, an attacker could theoretically connect a rogue switch and broadcast superior Bridge Protocol Data Units (BPDUs) to hijack the Root Bridge role, allowing them to intercept traffic. To mitigate this, **BPDU Guard** was enabled on all access ports. Additionally, the Main_MLS was hardcoded with a priority of 24576 to guarantee its role as the Root Bridge.

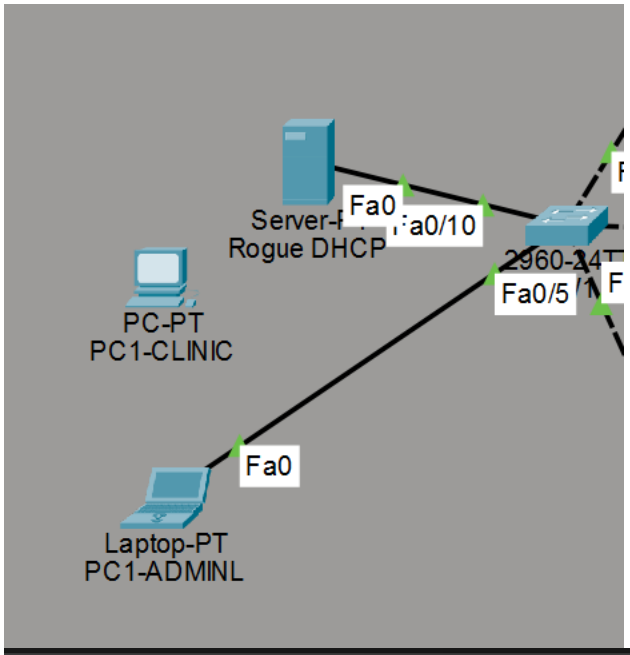
Proof of Concept: A rogue switch was connected to an access port on SW1. Upon receiving the first BPDU, the switch immediately transitioned the port into an err-disable state, protecting the topology.

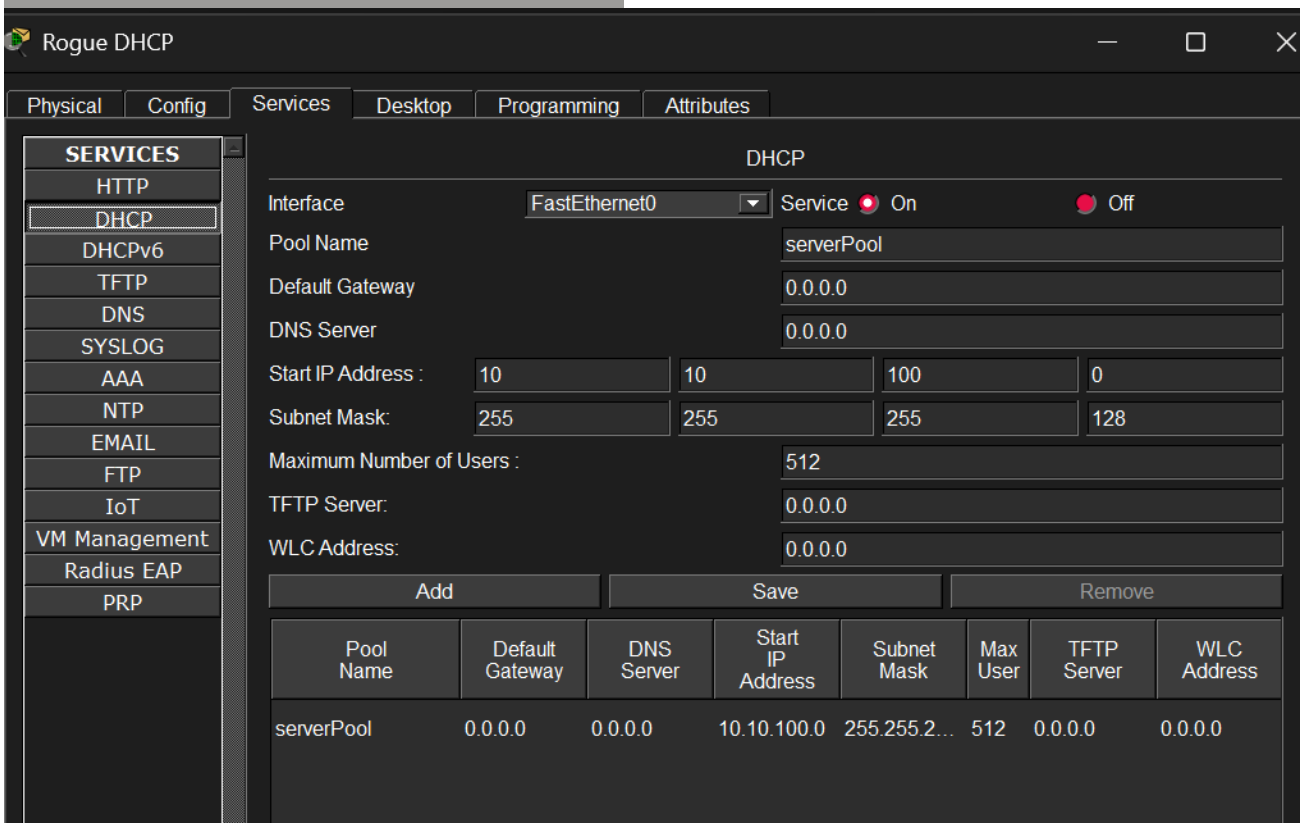



```
SW1#show interfaces fa0/18 status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/18                       disabled 1          auto    auto  10/100BaseTX

SW1#
```

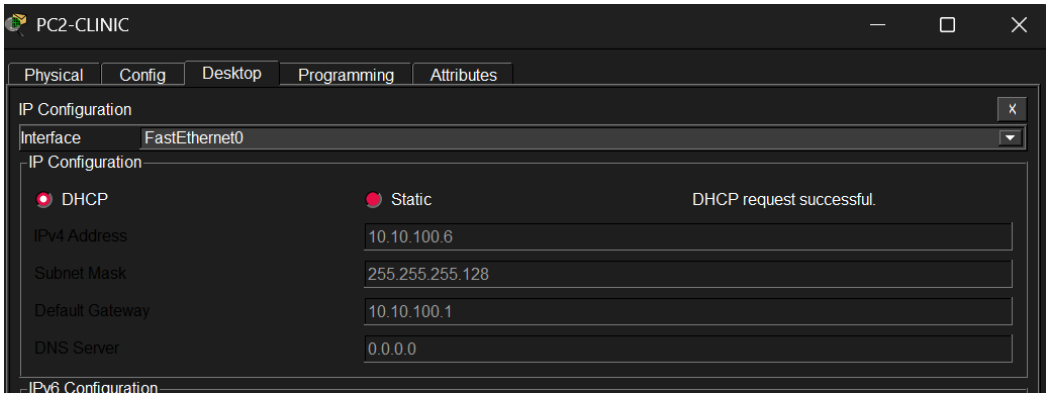
4.3. Preventing DHCP Starvation and ARP Spoofing

To prevent Man-in-the-Middle (MitM) attacks, **DHCP Snooping** was globally enabled. Only the trunk ports connecting to the Main_MLS were configured as trusted to distribute IP addresses. All other ports drop rogue DHCP offers. Furthermore, **Dynamic ARP Inspection (DAI)** was enabled, which uses the DHCP Snooping binding database to inspect all ARP packets, discarding invalid MAC-to-IP mappings.





SET UP FOR ROGUE DHCP SERVER



```
SW1>enable
Password:
SW1#show ip dhcp snooping binding
MacAddress          IpAddress        Lease(sec)  Type            VLAN  Interface
------------------  ---------------  ----------  -------------   ----  -----------------
00:E0:F9:AD:E9:4B   10.10.200.7      0           dhcp-snooping   200   FastEthernet0/5
Total number of bindings: 1
SW1#
```

4.4. Port Isolation (Blackhole VLAN)

As a fundamental physical security measure, all unused ports across all switches (Central and Branches) were assigned to a non-routable VLAN (VLAN 67) and administratively shut down. This prevents unauthorized network access even if a malicious actor gains physical access to a wall jack.

```
SW3#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
67   BlackHole                        active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
100  Clinical                         active    Fa0/3
200  Admin                            active    Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW3#
```

5. Security and Routing Validation

5.1. Secure Remote Administration Validation

To satisfy Requirement 4, Role-Based Access Control (RBAC) was implemented via SSH.
Proof of Concept: A successful remote connection was established from a clinical PC to the Main_MLS using the limited labuser account. Upon attempting to enter configuration mode, the user was required to elevate privileges using the enable password.

PC1-CLINIC

Physical | Config | Desktop | Programming | Attributes

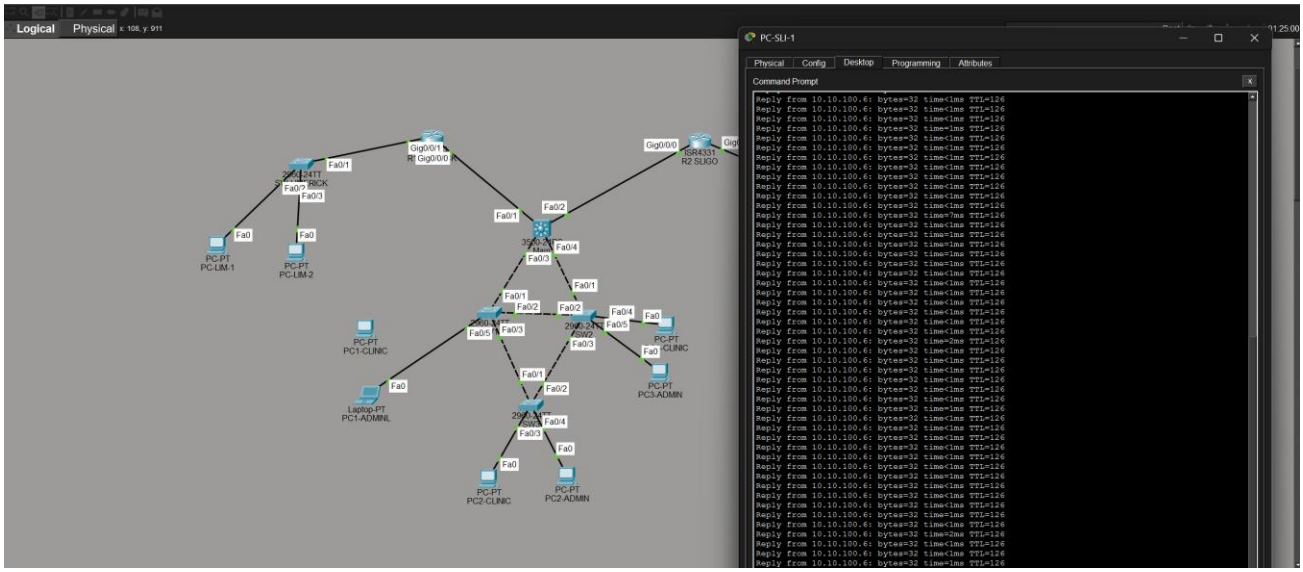
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l labuser 10.10.100.1

Password:



Main_MLS#show privilege
Current privilege level is 5
Main_MLS#enable
Password:
Main_MLS#show privilege
Current privilege level is 15
Main_MLS#
```

5.2. Network Reliability and Failover Validation

To guarantee uninterrupted healthcare services, network resilience was tested by simulating a catastrophic WAN failure at the Sligo clinic.

Proof of Concept: A continuous ping was initiated from Sligo to the Dublin campus. The primary Gigabit link (Gig0/0/0 on R2_Sligo) was manually shut down. The floating static route (via the Serial 0/1/0 interface to Limerick) successfully took over, and connectivity was restored within seconds.




```
R2_Sligo>enable
Password:
R2_Sligo#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 4 masks
C       10.0.0.4/30 is directly connected, GigabitEthernet0/0/0
L       10.0.0.6/32 is directly connected, GigabitEthernet0/0/0
S       10.10.100.0/25 [1/0] via 10.0.0.5
S       10.10.200.0/27 [1/0] via 10.0.0.5
     172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
S       172.16.10.0/27 [1/0] via 10.0.0.5
C       172.16.20.0/27 is directly connected, GigabitEthernet0/0/1
L       172.16.20.1/32 is directly connected, GigabitEthernet0/0/1

R2_Sligo#
```

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6. Scalability and Reliability Reflection

The Aurora Healthcare Group network prototype demonstrates a highly secure, reliable, and scalable architecture.

Scalability: By utilizing a Multilayer Switch at the core rather than a traditional Router-on-a-Stick, the network can easily scale to accommodate hundreds of new clinical staff members. The inter-VLAN routing is handled by hardware ASICs at wire speed, preventing the core from becoming a bottleneck. Furthermore, the IP subnetting scheme uses **CIDR (/25 and /27)**, leaving ample address space for future VLANs or clinic expansion.

Reliability: The implementation of a triangular topology ensures that a single point of failure (such as a fiber cut between Dublin and Sligo) does not isolate a clinic. The automatic failover provided by floating static routes guarantees that critical healthcare applications remain online. Decentralizing the DHCP services to the branch edge routers ensures that even during a total WAN isolation, local branch operations can continue without IP address conflicts.

Security: Security has been integrated at every layer. By combining strict Layer 2 access controls (Port Security, BPDU Guard, DAI) with administrative hardening (SSH, encrypted passwords, Blackhole VLANs), the network is robustly protected against both external intrusions and internal misconfigurations. This comprehensive approach ensures the confidentiality and integrity of Aurora Healthcare Group's critical data.